MILITARY SPECIFICATION

RESISTOR NETWORK, FIXED, FILM, ESTABLISHED RELIABILITY, STYLE RZR21, 16 PIN DIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

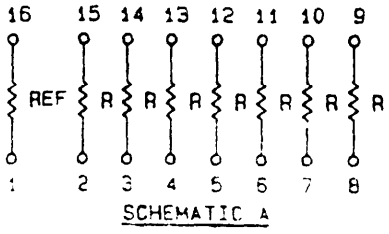
1.1 Scope. This specification covers the detail requirements for style RZR21, 16 pin dual in-line package (DIP) resistor network. This style is available in all characteristics and resistance tolerances.

1.2 Part or Identifying Number (PIN). Resistor networks covered by this specification shall be identified by a PIN which shall consist of the basic number of this specification and a coded number. The PIN shall be in the following form:

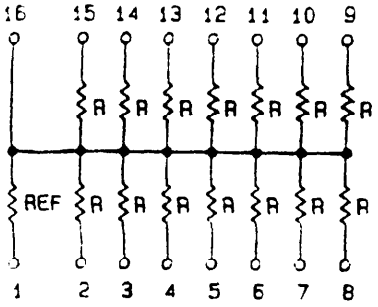
M874 — Military specification number

H12F1002AS — Coded number (See MIL-R-874)

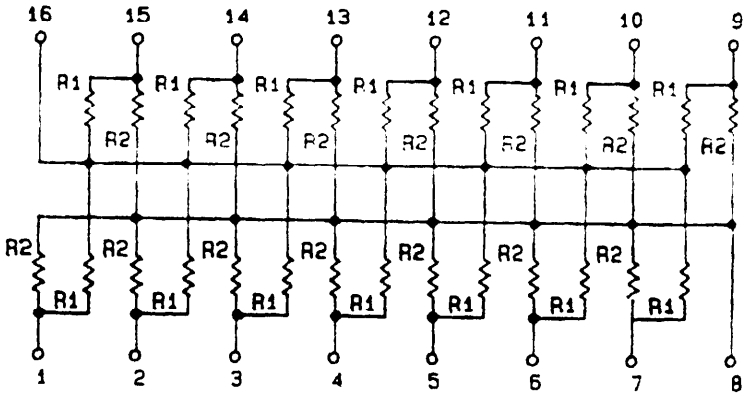
1.2.1 Schematics. The schematic of the resistor network shall be identified by a single letter in accordance with the drawings below. The resistor element R_{ref} shall be the reference resistor element used in determining the ratio accuracy (when applicable).



SCHEMATIC A



SCHEMATIC B



SCHEMATIC J

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.1 Standard resistance values. The standard resistance values and the resistance value designators for the schematic "J" are as specified in table I.

TABLE I. Standard resistance values.

Resistance designator	R1 (ohms)	R2 (ohms)	Resistance designator	R1 (ohms)	R2 (ohms)
A001	82	130	A010	330	470
A002	120	200	A011	330	680
A003	130	210	A012	1.5 k	3.3 k
A004	160	260	A013	3.0 k	6.2 k
A005	180	240	A014	180	270
A006	180	390	A015	270	270
A007	220	270	A016	560	560
A008	220	330	A017	560	1.2 k
A009	330	390	A018	620	2.7 k

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-874 - Resistor Networks, Fixed, Film, Established Reliability, General Specification for.

MIL-R-39032 - Resistors, Packaging of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-874, and as specified herein.

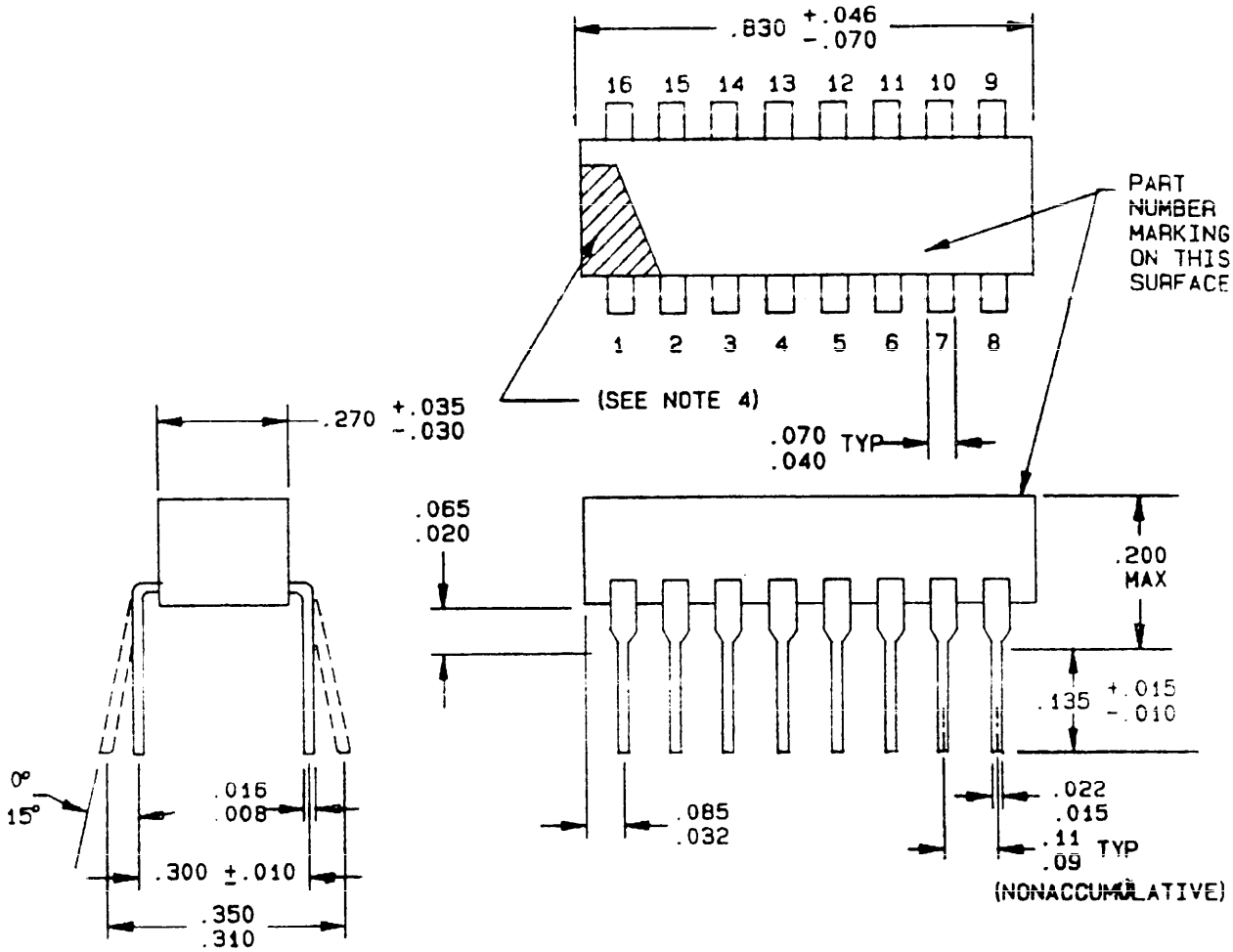
3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions as specified on figure 1.

3.2.1 Internal connection. Internal connections shall be accomplished by thermal compression bonding of gold to gold, to form a solid phase weld from the lead to the internal resistor connection. No solder connection eutetic or noneutetic shall be permitted in this construction.

3.3 Power rating. The power rating shall be as specified in table II, based on a full load operation at an ambient temperature of +70°C. For operation at temperatures higher than +70°C, derate in accordance with MIL-R-874.

3.4 Power conditioning. The power applied for power conditioning for schematics "A", "B", and "J" shall be 1.5 times rated power.

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Inches	mm	Inches	mm
.008	0.20	.070	1.78
.010	0.25	.085	2.16
.015	0.38	.090	2.29
.016	0.41	.110	2.79
.020	0.51	.135	3.43
.022	0.56	.200	5.08
.030	0.76	.270	6.86
.032	0.81	.300	7.62
.035	0.89	.310	7.87
.040	1.02	.350	8.89
.046	1.17	.830	21.08
.065	1.65		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The picturization of the styles above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.
4. Pin 1 locator shall be a dot, notch, stripe, or numeral 1 adjacent to pin number 1, in the shaded area.

FIGURE 1. RZR21 style resistors.

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3.5 Voltage rating. The maximum continuous working voltage for each style shall not exceed 100 volts dc or ac rms.

3.6 Resistance. Minimum and maximum resistance values shall be as specified in table III.

TABLE II. Power ratings

Schematic	Characteristic	Element power rating (watts)	Network power rating (watts)
A	H	0.2	1.6
	K	0.2	1.6
	M	0.2	1.6
B	H	0.1	1.5
	K	0.1	1.5
	M	0.1	1.5
J	H	0.050	1.4
	K	0.050	1.4
	M	0.050	1.4

TABLE III. Resistance values.

Resistance tolerance	Characteristic	Minimum resistance (ohms)	Maximum resistance (megohm)
B (±0.1 percent)	H	100	1
D (±0.5 percent)	H, K	100	1
F (±1.0 percent)	H, K, M	10.0	1
G (±2.0 percent)	K, M	10.0	1
J (±5.0 percent)	K, M	10.0	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-874.

4.2 "J" schematic tests. For the tests on schematic "J", the dc resistance measurements, resistance temperature characteristic measurement, and short time overload power application shall be made between each terminal and the applicable common terminal without compensation for the shunt circuits. The pin to pin dc resistance measurements shall be used directly to determine the acceptability to section 3 requirements.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Notes. The notes specified in MIL-R-874 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 PIN. PIN is a new term encompassing previous terms used in specifications such as part number, type designator, identification number, etc. (see 1.2).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1261-2)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-874/12	26 March 1992

3. DOCUMENT TITLE
RESISTOR NETWORK, FIXED, FILM, 16 PIN DIP, STYLE R2R21

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commander LABCOM	(1) Commercial (908) 544-3441	(2) AUTOVON 995-3441
c. ADDRESS (Include Zip Code) ATTN: SLCET-RS Fort Monmouth, NJ 07703	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete* 198-290

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